

English	(1001CJA6211200011)	*1001CJA6211200011*	Test Pattern
	 ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)	CLASSROOM CONTACT PROGRAMME (Academic Session : 2020 - 2021)	JEE(Advanced)
			UNIT TEST
			23-08-2020

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

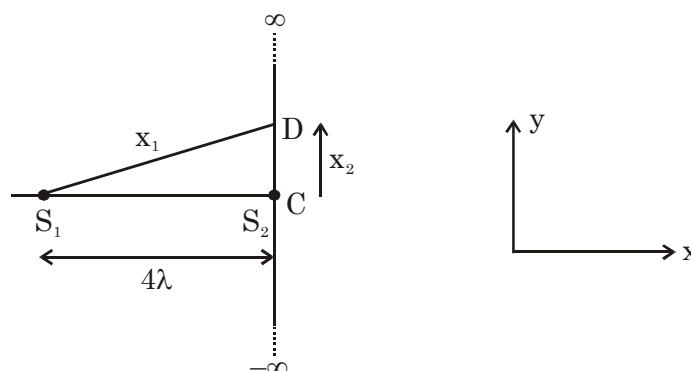
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

- Two coherent sound sources S_1 & S_2 are emitting sound waves in same phase as shown in figure.



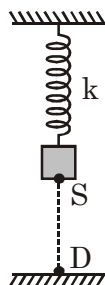
A detector D starts moving along y -axis from point C towards ∞ and it will detect maximas and minimas successively. Here $\Delta x = (x_1 - x_2)$ and taking maxima at C as central maxima, then :

- For first maxima $\Delta x = n\lambda$ where $n = 1$
- For 2nd minima $\Delta x = (n - 1/2)\lambda$ where $n = 2$
- For 3rd maxima $\Delta x = n\lambda$ where $n = 1$
- For 3rd minima $\Delta x = (n - 1/2)\lambda$ where $n = 2$

Space for Rough Work

2. A block is attached with a spring which is in its natural length initially as shown in figure. A source of sound which is emitting sound of frequency f is also attached with the block and system is released to fall from rest. Combined mass of block and source is m . A detector is placed directly below the source on ground which is at sufficient distance from spring block system. Then frequency registered by detector corresponding to waves emitted by source are

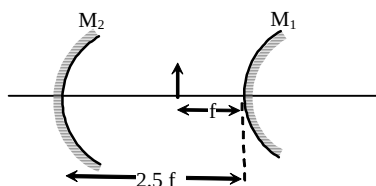
f_1 and f_2 respectively when block is at upper extreme and at mean position, then : ($\omega = \sqrt{\frac{k}{m}}$)



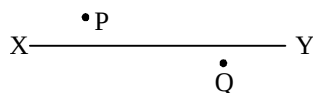
- (A) f_1 is $\frac{kfc}{kc + fmg(1 - \cos \omega/f)}$ (B) f_1 is $\frac{kfc}{kc - fmg(1 + \cos \omega/f)}$
- (C) f_2 is $\frac{kfc}{kc - fmg(\sin \omega/f)}$ (D) f_2 is $\frac{kfc}{kc - mg\omega}$
3. A transverse wave of wavelength 8 cm is moving along x-axis. There is also a prism having prism angle equal to phase difference between points (2, 0, 4) and (3, -1, 6) (in cm) due to propagation of wave what can be refractive index of prism so that all light rays can emerge through prism?
- (A) 2 (B) 1.2 (C) 1 (D) 1.3

Space for Rough Work

4. Out of the following, which statements are **CORRECT**?
- (A) Two plane mirrors are inclined to each other at an angle of 60° . If a ray of light incident on the first mirror is parallel to the second mirror, it is reflected from the second mirror parallel to the first mirror
- (B) A bird flying high up in the air does not cast a shadow on the ground because layers of atmosphere are dense
- (C) If a ray reflected successively from two plane mirrors inclined at a certain angle undergoes a deviation of 300° , then the number of images observable is 11
- (D) A clock indicates a time of 3:25. On seeing it in a plane mirror, the time appears as 8:35
5. The two mirrors M_1 & M_2 of focal length f each are placed coaxially as shown. Consider first reflection from M_1 & second from M_2 . Then which of the following are **CORRECT** :



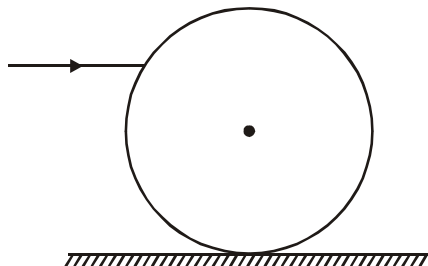
- (A) The image after two reflection coincides with object
- (B) Image after two reflection is inverted and its size is one fourth of object
- (C) Image after two reflection is real.
- (D) image after two reflection is virtual
6. Two points P and Q lie on either side of an axis XY as shown in the figure. It is desired to produce an image of luminous point object kept at P at Q using a spherical mirror, with XY as the optic axis. The mirror must be -



- (A) converging
- (B) diverging
- (C) positioned to the left of P
- (D) positioned to the right of Q

Space for Rough Work

7. A cylindrical glass rod of radius 0.1 m and refractive index $\sqrt{3}$ lies on a horizontal plane mirror. A horizontal ray of light moving perpendicular to the axis of the rod is incident on it at height H from the mirror so that it leaves the rod at a height of 0.1 m above the plane mirror. At centre to centre distance of d, a second similar rod, parallel to the first, is placed on the mirror, such that the emergent ray from the second rod is in line with the incident ray on the first rod, then



- (A) value of H can be 0.186 m (B) value of H can be 0.086 m
(C) value of d can be 0.32 m (D) value of d can be 0.12 m
8. A train of length 2π km is moving with a constant speed 36 kph along a circular track of radius 4 km. Engine of the train emits a whistle of frequency 400 Hz. Guard at the rear end of train hears this sound. (Speed of sound in 310 m/s)
- (A) Time taken to hear the sound by guard is $\frac{25\pi}{4}$ sec
(B) Time taken to hear the sound by guard is $\frac{5\pi}{4}$ sec
(C) Angle rotated by train during this time is $\frac{\pi}{64}$ rad
(D) Frequency heard by guard is approximately 400 Hz

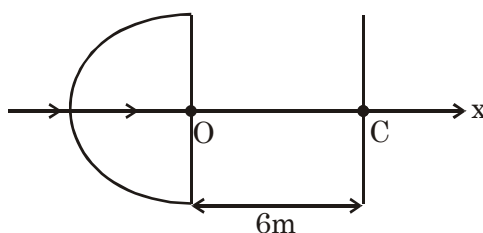
Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

9. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Figure shows a solid transparent semi cylinder placed at a distance of 6m from a screen. A narrow beam is incident along x-axis at O. If cylinder starts rotating about O in clockwise direction with angular speed ω then spot formed on screen starts moving upward. (Refractive index of cylinder is $\sqrt{2}$)



In List-I some quantities are given and in list-II their values in S.I. unit are given with suitable approximation

List-I	List-II
(I) Initial angular velocity of ray refracting from cylinder	(P) 6.0
(II) Angular velocity of ray refracting from cylinder at $t = \frac{\pi}{36}$ sec.	(Q) 4.4
(III) Time at which light spot will disappear on screen	(R) 0.1
(IV) Distance of spot from C when it disappear	(S) 2.5
	(T) 3.1
	(U) 8.3

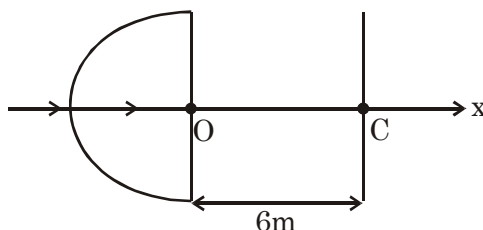
If $\omega = 6$ rad/s, then correct match will be

- (A) I $\rightarrow$ S, II $\rightarrow$ Q, III $\rightarrow$ P, IV $\rightarrow$ R (B) I $\rightarrow$ S, II $\rightarrow$ R, III $\rightarrow$ Q, IV $\rightarrow$ P
 (C) I $\rightarrow$ S, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ P (D) I $\rightarrow$ R, II $\rightarrow$ Q, III $\rightarrow$ P, IV $\rightarrow$ S

Space for Rough Work

10. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Figure shows a solid transparent semi cylinder placed at a distance of 6m from a screen. A narrow beam is incident along x-axis at O. If cylinder starts rotating about O in clockwise direction with angular speed ω then spot formed on screen starts moving upward. (Refractive index of cylinder is $\sqrt{2}$)



In List-I some quantities are given and in list-II their values in S.I. unit are given rounded off to two digits

List-I	List-II
(I) Initial angular velocity of ray refracting from cylinder	(P) 6.0
(II) Angular velocity of ray refracting from cylinder at $t = \frac{\pi}{36}$ sec.	(Q) 4.4
(III) Time at which light spot will disappear on screen	(R) 0.1
(IV) Distance of spot from C when it disappear	(S) 2.5
	(T) 3.1
	(U) 8.3

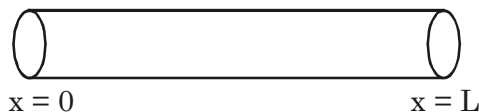
If $\omega = \frac{37}{5}$ rad/s, then correct match will be (use : $\sqrt{\frac{2}{7}} = 0.53$)

- | | |
|--|--|
| (A) I $\rightarrow$ S, II $\rightarrow$ U, III $\rightarrow$ P, IV $\rightarrow$ R | (B) I $\rightarrow$ T, II $\rightarrow$ U, III $\rightarrow$ R, IV $\rightarrow$ P |
| (C) I $\rightarrow$ T, II $\rightarrow$ U, III $\rightarrow$ P, IV $\rightarrow$ R | (D) I $\rightarrow$ U, II $\rightarrow$ T, III $\rightarrow$ P, IV $\rightarrow$ Q |

Space for Rough Work

11. Answer the following by appropriately matching the lists based on the information given in the paragraph.

An organ pipe of length ℓ is open at both ends and air inside it has density ρ . Equation of standing wave inside it is given as $S = (A \cos kx) \sin \omega t$.



Pipe is oscillating in 3rd harmonic. In list-I position of difference points is given and in list-II value of some quantity is given.

List-I

- (I) $x = \frac{\ell}{6}$
 (II) $x = \frac{2\ell}{3}$
 (III) $x = \frac{5\ell}{18}$
 (IV) $x = \frac{\ell}{9}$

List-II

- (P) $\frac{1}{16} \rho A^2 \omega^2$
 (Q) $\frac{3}{8} \rho A^2 \omega^2$
 (R) 0
 (S) $\frac{1}{4} \rho A^2 \omega^2$
 (T) $\frac{1}{8} \rho A^2 \omega^2$
 (U) $\frac{3}{16} \rho A^2 \omega^2$

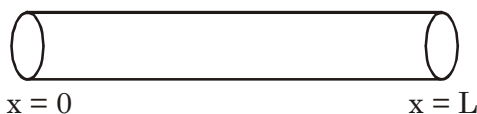
If list-II gives value of average potential energy density (potential energy per unit volume) corresponding to points given in list-I, then :

- (A) I $\rightarrow$ S, II $\rightarrow$ T, III $\rightarrow$ Q, IV $\rightarrow$ R
 (B) I $\rightarrow$ P, II $\rightarrow$ R, III $\rightarrow$ T, IV $\rightarrow$ Q
 (C) I $\rightarrow$ R, II $\rightarrow$ S, III $\rightarrow$ Q, IV $\rightarrow$ T
 (D) I $\rightarrow$ S, II $\rightarrow$ R, III $\rightarrow$ P, IV $\rightarrow$ U

Space for Rough Work

12. Answer the following by appropriately matching the lists based on the information given in the paragraph.

An organ pipe of length ℓ is open at both ends and air inside it has density ρ . Equation of standing wave inside it is given as $S = (A \cos kx) \sin \omega t$.



Pipe is oscillating in 3rd harmonic. In list-I position of difference points is given and in list-II value of some quantity is given.

List-I

- (I) $x = \frac{\ell}{6}$
 (II) $x = \frac{2\ell}{3}$
 (III) $x = \frac{5\ell}{18}$
 (IV) $x = \frac{\ell}{9}$

List-II

- (P) $\frac{1}{16} \rho A^2 \omega^2$
 (Q) $\frac{3}{8} \rho A^2 \omega^2$
 (R) 0
 (S) $\frac{1}{4} \rho A^2 \omega^2$
 (T) $\frac{1}{8} \rho A^2 \omega^2$
 (U) $\frac{3}{16} \rho A^2 \omega^2$

If list-II gives value of average kinetic energy density (Average K.E. per unit volume) corresponding to positions given in List-I, then :

- (A) I $\rightarrow$ R, II $\rightarrow$ S, III $\rightarrow$ U, IV $\rightarrow$ P
 (B) I $\rightarrow$ R, II $\rightarrow$ U, III $\rightarrow$ Q, IV $\rightarrow$ P
 (C) I $\rightarrow$ S, II $\rightarrow$ R, III $\rightarrow$ U, IV $\rightarrow$ P
 (D) I $\rightarrow$ U, II $\rightarrow$ S, III $\rightarrow$ Q, IV $\rightarrow$ P

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

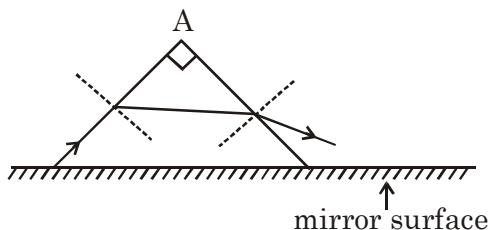
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

		+			
●	●	0	0	0	0
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	●
6	6	6	6	6	6
7	7	●	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

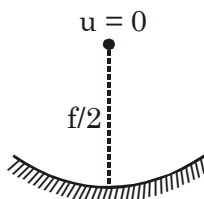
		-			
●	●	0	0	●	●
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	●	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A light ray grazes the right angled isosceles prism as shown and suffers deviation of 45° . If light ray is passed in such a way that it suffers minimum deviation then what will be its total deviation in degree after refraction and reflection from horizontal mirrored surface.



2. A particle is dropped along the axis from a height $f/2$ on a concave mirror of focal length f as shown in the figure. Find maximum speed (m/s) of image in subsequent motion of particle ($f = 40/3$ cm)



Space for Rough Work

3. A motorcyclist moving parallel to a long wall with speed 72 km/h blows the horn and hears echo of horn after 0.5 sec. Find distance between his line of motion and wall in meter. Speed of sound in air is 340 m/s.
4. A fish rising vertically to the surface of water in a lake uniformly at the rate of 3 m/s observes a kingfisher (bird) diving vertically towards the water at a rate of 9 m/s vertically above it. If the refractive index of water is $(4/3)$, find the actual velocity (in m/s) of the dive of the bird.
5. A tuning fork of frequency 480 Hz resonates with a tube closed at one end of length 16 cm and diameter 5 cm in fundamental mode. Calculate velocity of sound in air (in m/s).
6. Standing waves are produced by the superposition of two waves of the form
 $y_1 = 15 \sin (3\pi t - 5\pi x)$ cm;
 $y_2 = 15 \sin (3\pi t + 5\pi x)$ cm
 progressing along X-axis, where x is in cm and t in second. Find the speed (in cm/sec) of the particle at

$$t = 10 \frac{1}{9} \text{ s, } x = \frac{1}{3} \text{ cm.}$$

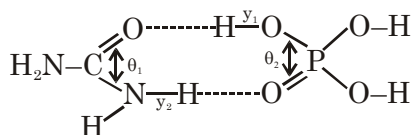
Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks: 32)

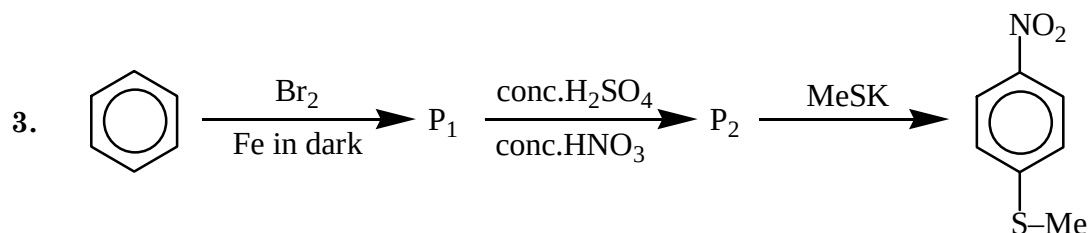
- This section contains **EIGHT** questions.
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- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. For the given compound select the **CORRECT** options :-



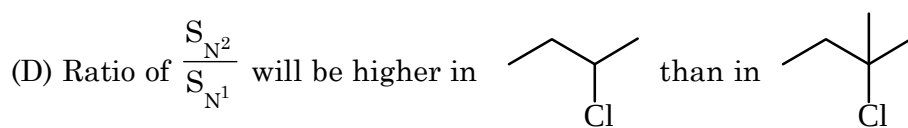
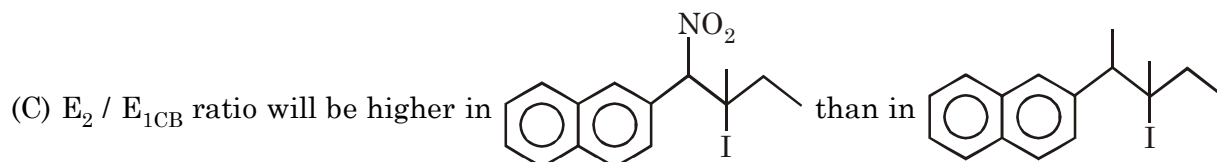
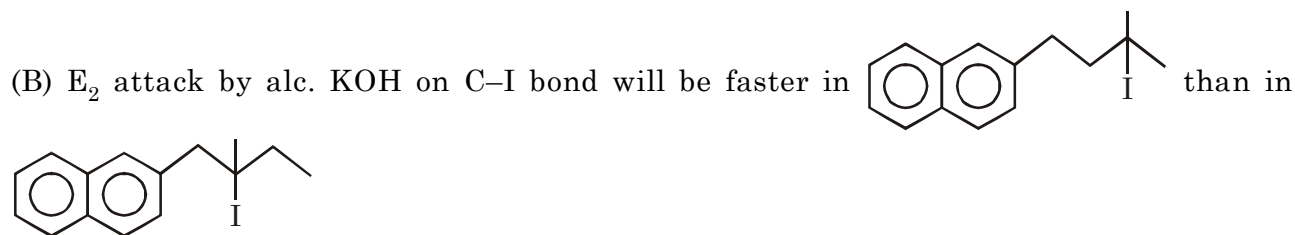
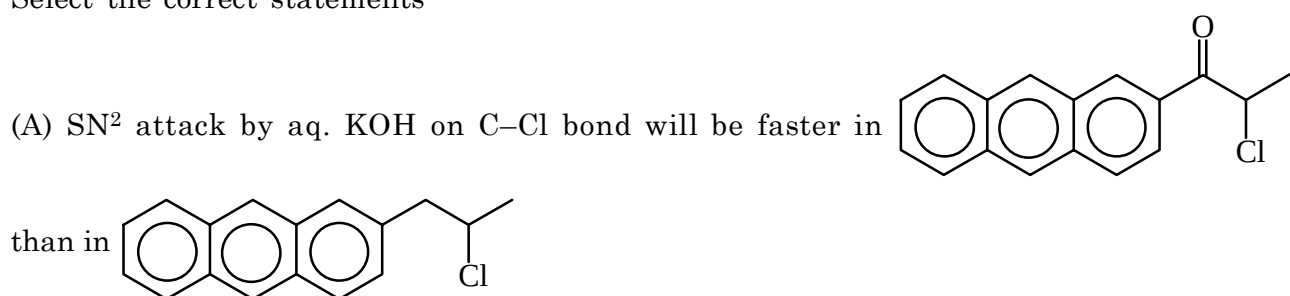
- (A) $\theta_1 < \theta_2$ (B) Intermolecular H-bonding is present
 (C) $y_1 < y_2$ (D) No $p_\pi-d_\pi$ bond present in the diagram
2. Select the **INCORRECT** statement(s) for $H_6P_4O_{13}$.
- (A) It contain four $p_\pi-d_\pi$ bonds (B) It contains P-P bonds
 (C) $\frac{\sigma \text{ bonds} + \text{lone pair}}{\pi \text{ bonds}} = 12$ (D) Cyclic structure

Space for Rough Work



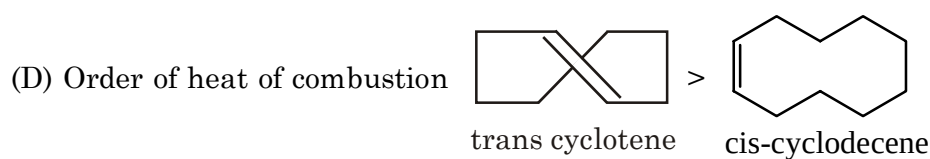
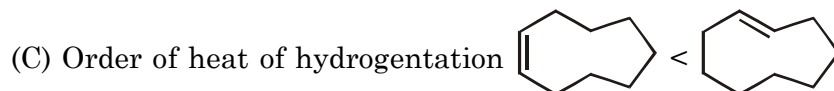
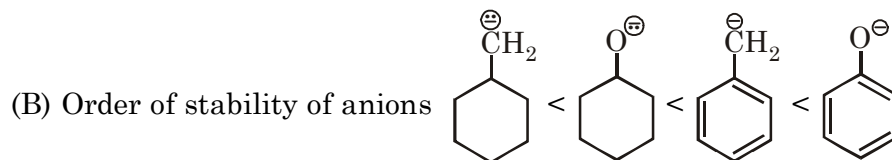
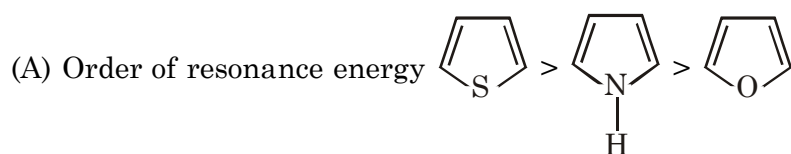
Select the correct statements about the given conversion :

- (A) P_1 gives white ppt. readily with silver nitrate solution
 (B) P_2 is formed by nucleophilic substitution reaction
 (C) P_1 is formed by electrophilic substitution reaction.
 (D) P_2 is formed by electrophilic substitution reaction.
4. Select the correct statements



Space for Rough Work

5. Select correct order among the following



Space for Rough Work

6. A mixture containing substance 'X' & 'Y' is mixed with a substance 'Z'. If 'Z' reacts with both X & Y, then, select the correct option(s)
- (A) Moles of X + moles of Y = Total moles of Z reacted "is always applicable"
- (B) Equivalents of X + equivalents of Y = Total equivalents of Z reacted "is always applicable"
- (C) $\left(\frac{\text{Equivalents of X}}{n\text{-factor of Z in reaction with X}} \right) + \left(\frac{\text{Equivalents of Y}}{n\text{-factor of Z in reaction with Y}} \right) = \text{Total moles of Z reacted}$ "is always applicable"
- (D) $\left(\frac{\text{Moles of Z reacted in reaction with X}}{\text{Moles of Z reacted in reaction with Y}} \right) = \text{Total moles of Z reacted}$ "is always applicable"
7. Which of the following statement (s) is /are true -
- (A) If H-atoms are in 3rd energy level, then number of maximum different spectral lines produced by 2 atoms will be equal to number of maximum different spectral lines produced by 3 H-atoms.
- (B) If H-atoms are in 3rd excited state, then number of maximum different spectral lines produced by 3 atoms will be equal to number of maximum spectral lines produced by 4 H-atoms.
- (C) Energy evolved from 3rd to 2nd energy level transition in He^+ cannot be used for any transition in Li^{2+}
- (D) Energy evolved by a He^+ ion which is in its 1st excited state can ionise a hydrogen atom which is in its ground state.
8. For an aqueous solution
- Information-I :** ratio of concentration of a solute in gm/litre to molarity is 25.
- Information-II :** Ratio of % w/w to % w/v is 0.8.
- Select correct statement -
- (A) Ratio of concentration of solute in gm/litre to % w/v is 10
- (B) Density of solution is 0.8 gm/ ml
- (C) If its molarity is 2M, molality will be $\frac{5}{3}$ m
- (D) Ratio of concentration of solute in % w/w to ppm is 100.

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

Answer the following by appropriately matching the lists based on the information given in the paragraph.

9. Match the list:

List- I
 (I) XeO_2F_2
 (II) IOF_5
 (III) SO_3

(IV) XeF_5^-

List- II
 (P) Both p_π - p_π and p_π - d_π bonds are present
 (Q) Non zero dipole moment
 (R) Maximum number of atoms are present in one plane
 (S) Minimum number of lone pair are present in molecule
 (T) Non planar
 (U) Can act as Lewis acid

Select the **CORRECT** match?

(A) I→R

(B) II→P

(C) III→U

(D) IV→T

Space for Rough Work

10. Match the list:

List- I

- (I) XeO_2F_2
 (II) IOF_5
 (III) SO_3
 (IV) XeF_5^-

List- II

- (P) Both $\text{p}_\pi\text{-p}_\pi$ and $\text{p}_\pi\text{-d}_\pi$ bonds are present
 (Q) Non zero dipole moment
 (R) Maximum number of atoms are present in one plane
 (S) Minimum number of lone pair are present in molecule
 (T) Non planar
 (U) Can act as Lewis acid

 Select **INCORRECT** match?

(A) III→P

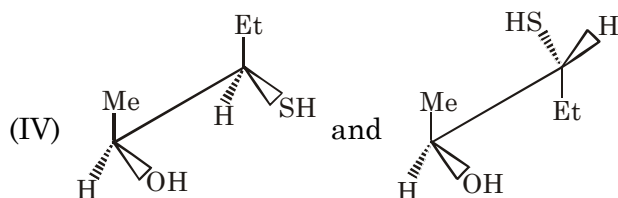
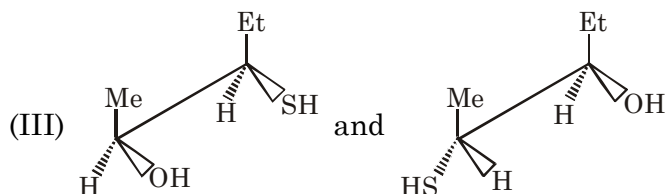
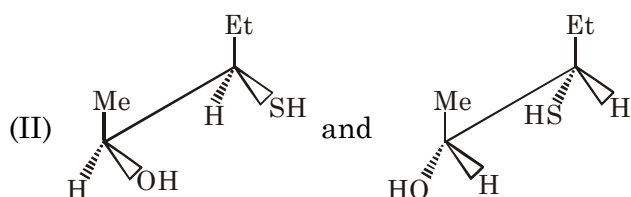
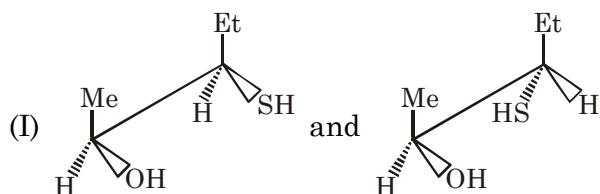
(B) IV→R

(C) I→Q

(D) II→S

Space for Rough Work

11. List - I
Pair of compounds



List- II
Statements related to pair of
compounds given in List-I

(P) They differ in the position of the
functional groups and hence are
positional isomers

(Q) They are mirror images of each other
and hence they are enantiomers

(R) They are optical isomers but are not
mirror images of each other and hence
are diastereomers

(S) They are geometrical isomers and are not
mirror images of each other and hence
diastereomers

(T) They belongs to same compound and
differ only in conformations.

(U) They differ in the parent carbon chain
and hence they are chain isomers.

Which of the following represent correct combinations from List-I and List-II

(A) I-P

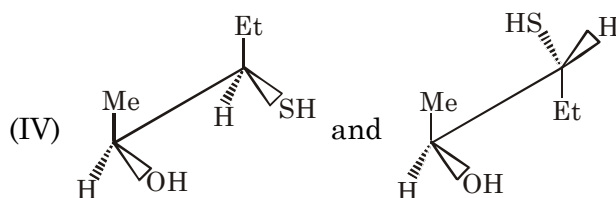
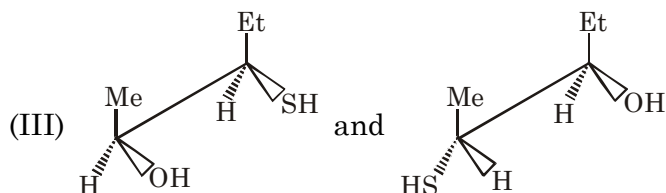
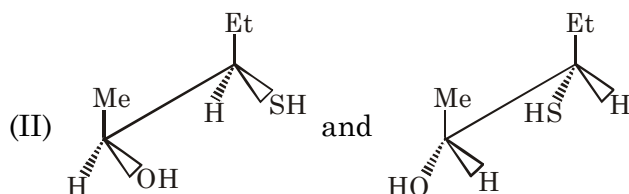
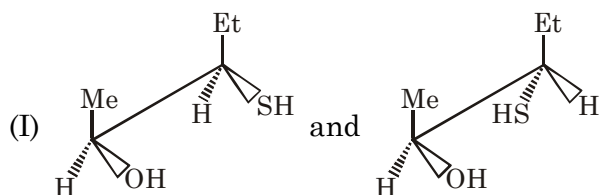
(B) II-Q

(C) III-S

(D) IV-U

Space for Rough Work

12. List - I
Pair of compounds



List- II
Statements related to pair of
compounds given in List-I

(P) They differ in the position of the
functional groups and hence are
positional isomers

(Q) They are mirror images of each other
and hence they are enantiomers

(R) They are optical isomers but are not
mirror images of each other and hence
are diastereomers

(S) They are geometrical isomers and are not
mirror images of each other and hence
diastereomers

(T) They belongs to same compound and
differ only in conformations.

(U) They differ in the parent carbon chain
and hence they are chain isomers.

Which of the following represent correct combinations from List-I and List-II

(A) I-R

(B) II-P

(C) III-U

(D) IV-S

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

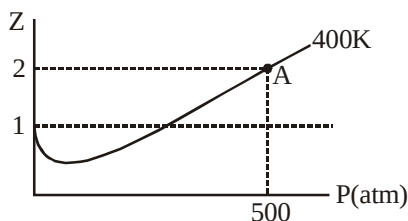
- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+										-									
●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
①	①	①	①	①	①	①	①	①	①	①	①	①	①	①	①	①	①	①	①
②	②	②	②	②	●	②	②	②	②	②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③	③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④	④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	●	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

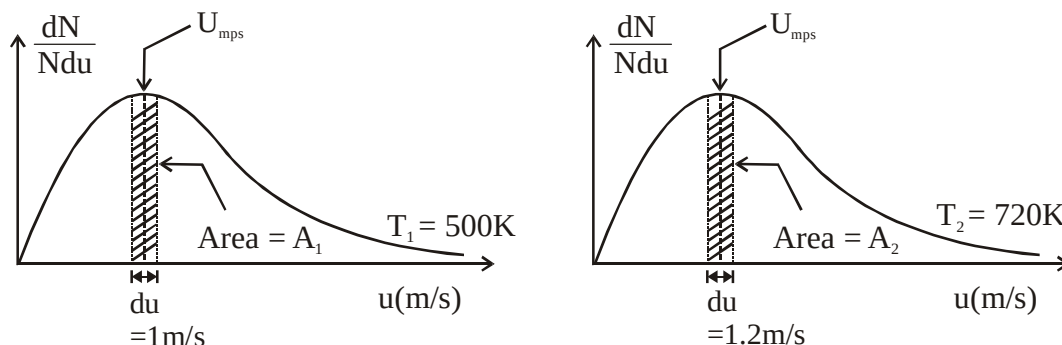
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- How many compounds of molecular formula C_5H_{10} are capable of showing optical isomerism?
- 'Z' v/s 'P' graph is plotted for 1 mole of hypothetical gas. Volume of gas (in litre) at this point 'A' is [R = 0.08 atm L/mole-K]



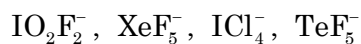
Space for Rough Work

3. For O_2 gas at T_1K and T_2K , following maxwell speed distribution is observed [not to the scale]
[$R = 8J/mol \cdot K$]



Ratio A_1 / A_2 is equal to

4. A hydrocarbon is burnt with just sufficient amount of oxygen. After combustion the volume contraction was 2.5 times of volume of hydrocarbon burnt. On passing this mixture through KOH, if volume contraction is twice of volume of hydrocarbon taken. Calculate no. of atoms in one molecule of the hydrocarbon.
5. MH_n is a polar molecule having planar geometry and 'M' belongs to 2nd period of modern periodic table then find maximum possible value of 'n'.
6. How many of the following planar species used atleast one axial d-orbitals in their hybridisation.



Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks: 32)**

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

-
1. The curve $xy = c$ ($c > 0$) and the circle $x^2 + y^2 = 1$ touch at two points, then -
(A) $c = 1/2$
(B) distance between points of contact is 2 unit
(C) slope of tangent at point of contact is -1
(D) length of latus rectum of $xy = c$ is 2
2. Let A(2, 0) and B(0, 2). Coordinates of point P on $2x + 3y + 1 = 0$ will be -
(A) (7, -5) if $|PA - PB|$ is maximum (B) $\left(-\frac{1}{5}, -\frac{1}{5}\right)$ if $|PA - PB|$ is maximum
(C) (7, -5) if $|PA - PB|$ is minimum (D) $\left(-\frac{1}{5}, -\frac{1}{5}\right)$ if $|PA - PB|$ is minimum
-

Space for Rough Work

3. Pair of lines through point P are tangents at Q and R to $y^2 = 4x$ such that normals at Q and R intersect at S, which lies on $y^2 = 4x$. If Q(1,2), then-
- (A) Circumcentre of triangle PQR is $\left(\frac{11}{2}, -\frac{3}{2}\right)$
- (B) Area of Quadrilateral PQRS is 16 unit²
- (C) Point S is (9, - 6)
- (D) Circumcentre of triangle QRS is $\left(\frac{11}{2}, -\frac{3}{2}\right)$
4. Tangents drawn to the ellipse E : $\frac{x^2}{24} + \frac{y^2}{16} = 1$ from the point P meets the co-ordinate axes at concyclic points, if the locus of P is the conic 'C', then
- (A) E & C intersect orthogonally.
- (B) shortest focal chord of conic C has length $4\sqrt{2}$
- (C) angle between two asymptotes of conic C is $\cos^{-1}0$
- (D) eccentricity of conic C is $\sqrt{\frac{3}{2}}$

Space for Rough Work

5. Let $x = (15 + \sqrt{224})^n = I + F$, where I & F are integral and fractional part of x respectively. Then (where $n \in \mathbb{N}$)
 (A) $x(1 - F) = 1$ (B) $x(1 - F) = 2^n$ (C) I is odd (D) I is even
6. In the expansion of $(2 - 3x)^{19}$, if at $x = 1$, r^{th} term is numerically greatest, then 'r' is
 (A) 12 (B) 13 (C) 14 (D) 17
7. If $r, k, p \in \mathbb{W}$, then $\sum_{r+k+p=10} {}^{30}C_r \cdot {}^{20}C_k \cdot {}^{10}C_p$ is equal to-
 (A) $\binom{60}{50}$ (B) $\binom{60}{30}$ (C) $\binom{60}{20}$ (D) $\binom{30}{10} \binom{30}{20}$
8. If ℓ is the length of the median through the vertex A of $\triangle ABC$, then $4\ell^2 =$
 (Symbols have their usual meaning)
 (A) $2b^2 + 2c^2 - a^2$ (B) $b^2 + c^2 + 2bc \cos A$
 (C) $a^2 + 4bc \cos A$ (D) $(a + 2b)^2 - 8s(s - c)$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

Answer the following by appropriately matching the lists based on the information given in the paragraph.

Consider a quadratic equation $f(x) = ax^2 + bx + c = 0$; $a, b, c \in \mathbb{R}$, $a \neq 0$ & $b^2 - 4ac > 0$.
 List-I puts some conditions on a, b & c . List-II gives location of roots of given quadratic equation.

List-I

List-II

(I) $\frac{-b}{2a} > 1$ and $af(1) > 0$

(P) both roots exceed 1

(II) $\frac{-b}{2a} < 1$ and $af(0) > 0$

(Q) atleast one root lies in $(1, \infty)$

(III) $\frac{-b}{2a} < 0$ and $af(1) < 0$

(R) atleast one root lies in $(-\infty, 1)$

(IV) $0 < \frac{-b}{2a} < 1$ and $af(1) < 0$

(S) atleast one root is negative

(T) both roots are negative

(U) both roots are less than 1

9. Which of the following options is the only **CORRECT** combination ?

(A) (I),(P)

(B) (I), (R)

(C) (II), (T)

(D) (II), (U)

Space for Rough Work

Answer the following by appropriately matching the lists based on the information given in the paragraph.

Consider a quadratic equation $f(x) = ax^2 + bx + c = 0$; $a, b, c \in \mathbb{R}$, $a \neq 0$ & $b^2 - 4ac > 0$. List-I puts some conditions on a, b & c . List-II gives location of roots of given quadratic equation.

List-I

List-II

(I) $\frac{-b}{2a} > 1$ and $af(1) > 0$

(P) both roots exceed 1

(II) $\frac{-b}{2a} < 1$ and $af(0) > 0$

(Q) atleast one root lies in $(1, \infty)$

(III) $\frac{-b}{2a} < 0$ and $af(1) < 0$

(R) atleast one root lies in $(-\infty, 1)$

(IV) $0 < \frac{-b}{2a} < 1$ and $af(1) < 0$

(S) atleast one root is negative

(T) both roots are negative

(U) both roots are less than 1

10. Which of the following options is the only **CORRECT** combination ?

(A) (III), (Q)

(B) (III), (U)

(C) (IV), (P)

(D) (IV), (S)

Space for Rough Work

Answer the following by appropriately matching the lists based on the information given in the paragraph.

List-I contains some trigonometric equations and List-II provides number of solutions in given interval.

List-I

(I) $2\cot 2x - 3\cot 3x = \tan 2x$

(II) $\sin^{10} x + \cos^{10} x = \frac{29}{16} \cos^4 2x$

(III) $\cos^4 2x + 2\sin^2 2x = 17(\cos x + \sin x)^8$

(IV) $\sin\left(x + \frac{\pi}{9}\right) = \cos\left(2x - \frac{59\pi}{180}\right)$

List-II

(P) 0 in $\left(0, \frac{\pi}{2}\right)$

(Q) 2 in $(0, \pi)$

(R) 4 in $(0, \pi)$

(S) 4 in $(0, 2\pi)$

(T) 2 in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(U) 2 in $\left(\frac{\pi}{2}, \pi\right)$

11. Which of the following options is the only **CORRECT** combination ?

- (A) (I),(Q) (B) (I), (T) (C) (II), (R) (D) (II), (S)

Answer the following by appropriately matching the lists based on the information given in the paragraph.

List-I contains some trigonometric equations and List-II provides number of solutions in given interval.

List-I

(I) $2\cot 2x - 3\cot 3x = \tan 2x$

(II) $\sin^{10} x + \cos^{10} x = \frac{29}{16} \cos^4 2x$

(III) $\cos^4 2x + 2\sin^2 2x = 17(\cos x + \sin x)^8$

(IV) $\sin\left(x + \frac{\pi}{9}\right) = \cos\left(2x - \frac{59\pi}{180}\right)$

List-II

(P) 0 in $\left(0, \frac{\pi}{2}\right)$

(Q) 2 in $(0, \pi)$

(R) 4 in $(0, \pi)$

(S) 4 in $(0, 2\pi)$

(T) 2 in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(U) 2 in $\left(\frac{\pi}{2}, \pi\right)$

12. Which of the following options is the only **INCORRECT** combination ?

- (A) (III), (P) (B) (III), (U) (C) (IV), (S) (D) (IV), (T)

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

		+			
●	●	0	0	0	0
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	●
6	6	6	6	6	6
7	7	●	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

		-			
●	●	0	0	●	●
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	●	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If circles C_1 & C_2 are drawn by taking foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ as centres, and a common tangent L is drawn to touch all three curves, then product of the radii of C_1 & C_2 is given by
2. If a , b , c are three complex numbers such that $a^2 + b^2 + c^2 = 0$ and

$$\begin{vmatrix} (b^2 + c^2) & ab & ac \\ ab & (c^2 + a^2) & bc \\ ac & bc & (a^2 + b^2) \end{vmatrix} = Ka^2b^2c^2, \text{ then value of } K \text{ is}$$

Space for Rough Work

3. Let A and B are two points on x-axis and y-axis respectively and circle passes through A, B and O (origin). Tangent at origin is at a distance of 3 unit and 5 unit from point A and B respectively, then radius of circle is
4. The sum of first 99 terms of the series $\frac{3}{1^2} + \frac{5}{1^2 + 2^2} + \frac{7}{1^2 + 2^2 + 3^2} + \dots$, is
5. If $A = \sum_{n=1}^8 \sqrt{3 + \sqrt{n}}$ and $B = \sum_{n=1}^8 \sqrt{3 - \sqrt{n}}$, then the value of $\frac{A}{B}$ is
6. $\frac{(\cot 10^\circ - \tan 10^\circ - 2\operatorname{cosec} 20^\circ)(\cot 10^\circ + \tan 10^\circ)}{(2 \cot 20^\circ - \cot 10^\circ - \tan 10^\circ)\operatorname{cosec} 20^\circ}$ is equal to

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	8	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-1 In all other cases	32
I(ii)	Matching Lists Type (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	12
II	Numerical Value Type (Up to second decimal place)	6	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	18

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator